

Both Tumor Depth and Diameter Are Predictive of Sentinel Lymph Node Status and Survival in Merkel Cell Carcinoma

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BACKGROUND: The purposes of this study were 1) to determine the impact of primary tumor-related factors on the prediction of the sentinel lymph node (SLN) status and 2) to identify clinical and pathologic factors associated with survival in Merkel cell carcinoma (MCC). **METHODS:** An institutional review board-approved, retrospective review of patients with MCC treated between 1988 and 2011 at a single center was performed. Patients were categorized into 5 groups: 1) negative SLN, 2) positive SLN, 3) clinically node-negative but SLN biopsy not performed, 4) regional nodal disease without a known primary tumor, and 5) primary MCC with synchronous clinically evident regional nodal disease. Factors predictive of the SLN status were analyzed with logistic regressions, and overall survival (OS) and disease-specific survival (DSS) were analyzed with Cox models and competing risk models assuming proportional hazards, respectively. **RESULTS:** Three hundred seventy-five patients were analyzed, and 70% were male; the median age was 75 years. The median tumor diameter was 1.5 cm (range, 0.2-12.5 cm), and the median tumor depth was 4.8 mm (range, 0.3-45.0 mm). One hundred ninety-one patients underwent SLN biopsy, and 59 (31%) were SLN-positive. Increasing primary tumor diameter and increasing tumor depth were associated with SLN positivity ($P=.007$ and $P=.017$, respectively). Age and sex were not associated with the SLN status. Immunosuppression, increasing tumor diameter, and increasing tumor depth were associated with worse OS ($P=.007$, $P=.003$, and $P=.025$, respectively). DSS differed significantly by group and was best for patients with a negative SLN and worst for those with primary MCC and synchronous clinically evident nodal disease ($P=.018$). **CONCLUSION:** For patients with MCC, increasing primary tumor diameter and increasing tumor depth are independently predictive of a positive SLN, worse OS, and worse DSS. Tumor depth should be routinely reported when primary MCC specimens are being evaluated histopathologically. *Cancer* 2015;121:3252-60. © 2015 American Cancer Society.

KEYWORDS: Merkel cell carcinoma, sentinel lymph node, survival, tumor depth, tumor diameter.

INTRODUCTION

Merkel cell carcinoma (MCC) was first described in 1972 by Toker,¹ who reported on 5 patients with unusual skin neoplasms with histologically anastomosing trabeculae and cell nests in the dermis. It is estimated that 1500 cases are diagnosed annually in the United States with an age-adjusted incidence of <0.5 cases per 100,000 person-years. Recently, there has been an increase in the incidence of MCC; this may be a result of increased awareness and improved diagnostic techniques (especially immunohistochemistry). Cytokeratin 20 is a highly specific histologic marker for MCC, and it has been available since 1992.² Cytokeratin 20, in conjunction with thyroid transcription factor 1, can reliably distinguish MCC from small cell lung carcinoma and other cutaneous malignancies. The aging population, increasing ultraviolet A and B exposure, and immunosuppression have been cited as potential causes of the increased incidence of MCC cases in the United States.³

Regional nodal involvement is frequent in MCC, with an estimated incidence of 24% to 31% at presentation.⁴⁻⁷ Sentinel lymph node (SLN) biopsy is increasingly used for the identification of occult nodal involvement in MCC.^{8,9} Although it is not universally accepted, most guidelines recommend SLN biopsy for patients with localized MCC to stage the clinically

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node-negative regional nodal basin and allow for early treatment of patients with occult nodal disease before it becomes clinically evident.¹⁰ Resection of the primary tumor and therapeutic lymph node dissection are recommended for patients presenting with clinically positive regional lymph node disease, and postoperative radiation is frequently used in patients with clinically node-positive nodal basins.¹¹ The management of SLN-positive MCC is less well defined, with completion lymph node dissection or regional nodal radiation both being advocated as a definitive treatment for the SLN-positive regional nodal basin.^{4,12}

Some of the factors that have been shown to influence the risk of SLN positivity and overall survival (OS) in MCC include the tumor diameter, presence of lymphovascular or angiolymphatic invasion, and markers of proliferation such as Ki-67.¹³ In contrast to melanoma, the impact of the depth of invasion (Breslow thickness) on SLN status and survival in MCC is undefined. Data on tumor depth, particularly with respect to the SLN status, have been limited because of the rarity of the disease and incomplete synoptic reporting of the tumor depth in pathology reports. We sought to evaluate clinicopathologic features, including the tumor depth and diameter,¹⁴⁻¹⁷ and their relation with SLN status and survival in patients with MCC.

MATERIALS AND METHODS

Patient Population

We performed an institutional review board–approved, single-institution, retrospective study of patients with a diagnosis of MCC who were evaluated between 1988 and 2011 at the Moffitt Cancer Center. Patient demographics and clinicopathologic features were determined from a prospectively collected MCC database including data from patient charts, pathology records, and the institutional tumor registry. Patients received pathological confirmation of MCC from dermatopathologists at the Moffitt Cancer Center. A pathology re-review of the entire patient cohort was not performed.

The tumor diameter was derived from clinic notes, operative notes, or pathology reports. The tumor depth was measured from the granular layer of the epidermis to the deepest level of invasion. Lymphovascular invasion (LVI) has only recently become part of the synoptic reporting at our center for MCC. Hence, there was an insufficient sample size to determine the impact of LVI on nodal status or survival.

Treatment decisions were made after discussions at a weekly multidisciplinary tumor board conference attended

by surgical oncologists, pathologists, medical oncologists, and radiation oncologists.

Radiation therapy and chemotherapy were used in select cases of MCC. These therapies were not consistently used in the patient population. Therefore, data on these 2 therapies were not included. The presence of immunosuppression was defined as the use of medications known to be immunosuppressants (eg, steroids) or the presence of comorbid medical conditions leading to immunosuppression (eg, human immunodeficiency virus disease).

Patients with stage I, II, or III disease were included in the current analysis. Patients with stage IV disease at presentation were excluded. Patients were categorized into 5 groups for the purposes of statistical analysis: 1) negative SLN, 2) positive SLN, 3) clinically node-negative but SLN biopsy not performed, 4) palpable regional nodal disease at presentation without a known primary tumor (Merkel cell carcinoma of unknown primary [MCCUP]), and 5) primary MCC with synchronous clinically positive regional nodal disease.

SLN biopsy was not performed for some patients with multiple comorbidities who were thought to be at higher risk for adverse events if general anesthesia was used and for patients in whom a SLN could not be identified on preoperative radionuclide lymphoscintigraphy. Ten patients were treated before 1988 (ie, before SLN biopsy became available). However, the majority of the patients who did not undergo SLN biopsy were in fact treated after 1988.

Statistical Analysis

OS was defined as the time from the date of diagnosis to the date of death for patients who were dead, and as the time from the date of diagnosis to the date on which they were last known to be alive for patients who were alive at last follow-up. Disease-specific survival (DSS) was defined as the time from diagnosis to MCC-related death; those alive at last follow-up were censored, and non-MCC-related deaths were treated as competing risks. Time to recurrence was defined as the time from surgical treatment (wide excision and SLN biopsy in clinically node-negative patients and therapeutic lymphadenectomy in patients presenting with palpable lymphadenopathy) to recurrence; those alive without recurrence at last follow-up were censored, and all deaths were treated as competing risks.

Univariate logistic regression was used to examine the association between clinical and pathologic variables and the SLN status. Tumor depth and diameter values were transformed with a $\log_2$ base because of their non-normal distribution.

TABLE 1. Patient and Pathologic Characteristics for 375 Patients With Merkel Cell Carcinoma Treated at the Moffitt Cancer Center Between 1988 and 2011

Variable	Value
Age at diagnosis, median (range), y	75 (30-96)
Sex, No. (%)	
Male	264 (70)
Female	111 (30)
Location, No. (%)	
Head and neck	168 (45)
Upper extremity	87 (23)
Lower extremity	49 (13)
Trunk	30 (8)
Unknown primary	41 (11)
Presence of immunosuppression, No. (%)	
Yes	55 (15)
No	315 (84)
Unknown	5 (1)
Presence of lymphovascular invasion, No. (%)	
Yes	15 (4)
No	40 (11)
Unknown	320 (85)
Tumor diameter, median (range), cm	1.5 (0.2-12.5)
Tumor depth, median (range), mm	4.8 (0.3-45.0)
Classification by nodal status, No. (%)	
Negative SLN	132 (35)
Positive SLN	59 (16)
No SLN biopsy performed	84 (22)
MCCUP	28 (7)
MCC+RLN	44 (12)
Unknown ^a	28 (7)

Abbreviations: MCC+RLN, primary Merkel cell carcinoma with synchronous clinically evident regional nodal disease; MCCUP, Merkel cell carcinoma of unknown primary; SLN, sentinel lymph node.

^aTwenty eight patients who were initially treated elsewhere had unknown nodal status

Univariate Cox proportional hazard models were used to assess the association between clinical factors and OS. Competing risk regression models assuming propor-

tional hazards were incorporated to evaluate factors associated with recurrence and DSS. Comparisons of OS between nodal groups were conducted with a log-rank test and Kaplan-Meier curves. Recurrence rates and DSS were compared between nodal groups with Gray's test and cumulative incidence curves.

RESULTS

Patient Demographics

A total of 375 MCC patients were analyzed. Table 1 summarizes the clinicopathologic features of the study population. The median age at diagnosis was 75 years (range, 30-96 years), and 264 patients (70%) were male. The most common location for MCC was the head and neck (45%), which was followed by the upper extremities (23%), lower extremities (13%), and trunk (8%). Two hundred seventy five patients presented with clinically node-negative disease, and 191 (50.9%) underwent SLN biopsy. Twenty-eight patients presented with MCCUP, and an additional 44 presented with a known primary and clinically positive nodal disease. Fifty-five patients (15%) had a documented history of immunosuppression. Four of the 28 patients with MCCUP were immunosuppressed.

Pathologic Characteristics

Data on the tumor diameter and depth were available for 206 and 188 patients, respectively, with 130 of these patients having both the depth and the diameter documented. The median tumor diameter was 1.5 cm (range, 0.2-12.5 cm), and the median tumor depth was 4.8 mm (range, 0.3-45 mm). The median follow-up time for all

TABLE 2. Univariate Analysis of Clinicopathologic Features Predictive of the SLN Status in 191 Patients Undergoing Sentinel Node Biopsy for Merkel Cell Carcinoma

Variable	Negative SLN	Positive SLN	Risk of Positive SLN		P
			Odds Ratio	95% Confidence Interval	
Sex, No.					.32
Female	43	15	1		
Male	89	44	1.4	0.7-2.8	
Presence of immunosuppression, No.					.69
No	115	51	1		
Yes	15	8	1.2	0.5-3.0	
Median tumor diameter, cm (n = 120) ^a	1.2	1.6	1.7	1.2-2.6	.007
Median tumor depth, mm (n = 139) ^a	4.4	5.5	1.4	1.1-1.9	.017
Location of primary, No.					.04
Head and neck	64	21	1		
Lower extremity	18	16	2.7	1.2-6.2	
Trunk	9	8	2.7	0.9-7.9	
Upper extremity	40	14	1.1	0.5-2.3	

Abbreviation: SLN, sentinel lymph node. Numbers may not add up due to cases with missing information.

^aA log₂ transformation (ie, doubling in depth or diameter) was used.

The number of cases may not add up to 191 in each subgroup due to some cases with missing information.

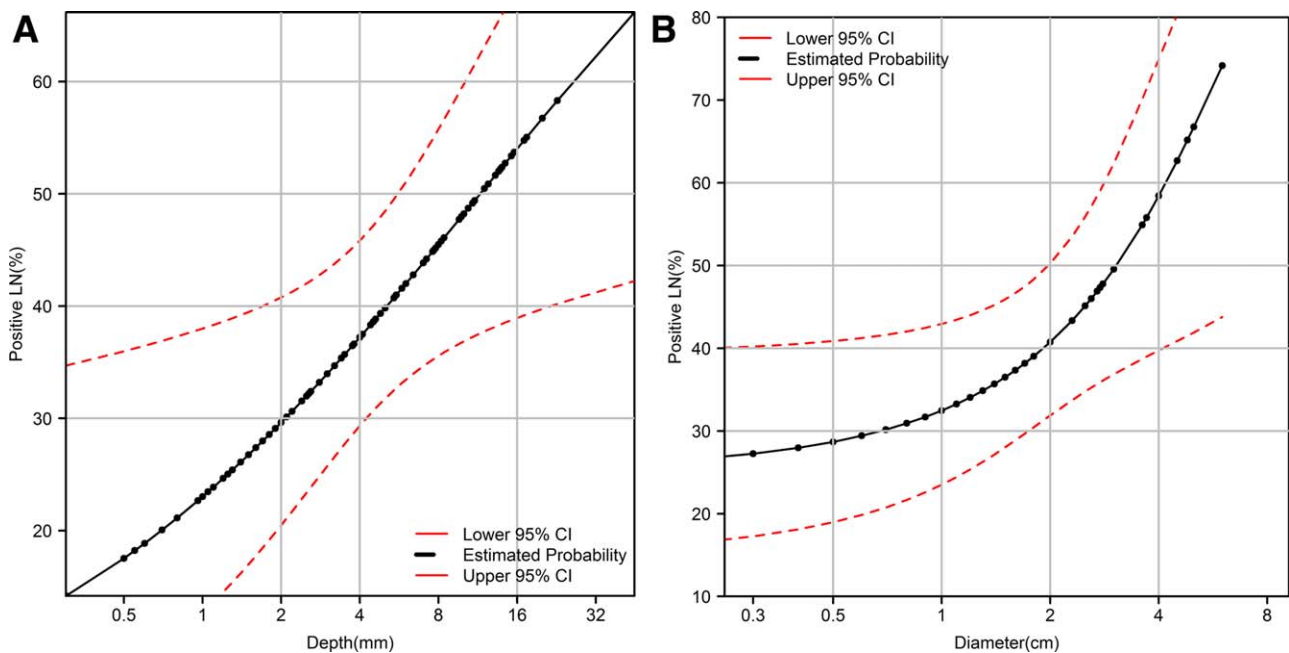


Figure 1. (A) Tumor depth versus the risk of a positive SLN in Merkel cell carcinoma. The estimated probability of having a positive SLN for a tumor 1 mm in depth was 23% (95% CI, 13%-38%), and the estimated probability of having a positive SLN for a tumor 8 mm in depth was 45% (95% CI, 36%-56%). (B) Tumor diameter versus the risk of a positive SLN in Merkel cell carcinoma. The estimated probability of having a positive SLN for a tumor 1 cm in diameter was 32% (95% CI, 23%-43%), and the estimated probability of having a positive SLN for a tumor 4 cm in diameter was 59% (95% CI, 42%-74%). CI indicates confidence interval; LN, lymph node; SLN, sentinel lymph node.

patients was 2.2 years (interquartile range, 1.1-5.2 years). The median follow-up time by nodal groups was 2.2 years for the SLN-negative group, 2.2 years for the SLN-positive group, 2.1 years for the clinically node-negative group for which SLN biopsy was not performed, 3.2 years for the MCCUP group, and 3.9 years for the group with primary MCC and synchronous clinically positive regional nodal disease.

Predictors of SLN Positivity

Fifty-nine of 191 patients (31%) had a positive SLN biopsy (Table 2). No significant association was demonstrated between either sex or age and SLN status. Patients with tumors located on the lower extremities and trunk had an increased risk of a positive SLN in comparison with other disease locations (Table 2). The presence of immunosuppression had no demonstrable impact on the likelihood of a positive SLN.

Among the 191 patients who underwent SLN biopsy, 91 (48%) had both depth and diameter reported. The tumor depth and the tumor diameter were moderately correlated (Pearson correlation, 0.53; $P < .0001$). Increasing tumor depth and increasing tumor diameter were each associated with a significantly increased likelihood of a positive SLN (Table 2 and Fig. 1). The odds of a positive

SLN increased 1.4 times (95% confidence interval [CI], 1.1-1.9) as the tumor depth doubled ($P = .017$). The odds of a positive SLN increased 1.7 times (95% CI, 1.2-2.6) as the tumor diameter doubled ($P = .007$).

Recurrence

Sex, increasing age, and the location of the primary tumor were not associated with recurrence. Increasing tumor diameter and increasing tumor depth were also not significantly associated with recurrence (Table 3).

There was no statistically significant difference in the risk of disease recurrence across the 5 nodal status groups identified with Gray's test with a competing risk approach ($P = .39$; Fig. 2A). However, there was a trend toward a lower recurrence risk in patients with a negative SLN versus a positive SLN (Gray's test, $P = .095$; Fig. 2B). The recurrence rate was lowest for patients who had a negative SLN biopsy, who were followed closely by MCCUP patients. Patients who had a positive SLN had the highest rate of recurrence.

OS

One hundred fifty-four patient deaths occurred during the study period; 70 patients (45%) were confirmed to have died of MCC, whereas 84 patients (55%) died of

TABLE 3. Univariate Analysis of Clinicopathological Features Versus Recurrence and Survival in Merkel Cell Carcinoma

Variable	Recurrence			Overall Survival			Disease-Specific Survival		
	HR	95% CI	<i>P</i> ^a	HR	95% CI	<i>P</i> ^a	HR	95% CI	<i>P</i> ^a
Sex (male versus female)	1.5	0.8-2.9	.25	1.2	0.8-1.7	.27	2.0	1.1-3.7	.03
Age (per 10-y increase)	1.0	0.8-1.3	.94	1.6	1.3-1.9	<.0001	1.5	1.1-1.9	.003
Presence of immunosuppression	1.4	0.7-2.9	.35	1.8	1.2-2.7	.007	1.4	0.7-2.6	.31
Tumor diameter (cm)	1.2	0.9-1.7	.28	1.4	1.1-1.7	.003	3.7	1.4-9.7	.008
Tumor depth (mm)	1.1	0.9-1.4	.35	1.3	1.0-1.6	.025	3.9	1.6-9.6	.003
Location of primary			.60			.15			.49
Head and neck	1			1			1		
Lower extremity	1.5	0.7-3.1		1.0	0.6-1.6		1.0	0.5-2.2	
Unknown	0.8	0.3-3.2		0.7	0.4-1.3		0.5	0.2-1.4	
Trunk	0.8	0.3-2.3		1.8	1.0-3.0		1.4	0.7-3.0	
Upper extremity	0.7	0.3-1.5		0.9	0.6-1.4		0.9	0.5-1.7	
Nodal group			.39			.007			.018
Negative SLN	1			1			1		
Positive SLN	2.1	0.9-4.6		1.5	0.9-2.5		2.1	1.0-4.4	
SLN biopsy not performed	2.3	1.1-4.9		2.0	1.3-2.9		2.5	1.3-4.9	
MCCUP	1.8	0.6-5.6		0.8	0.3-1.8		1.5	0.5-4.6	
MCC+RLN	2.0	0.8-5.0		1.9	1.2-3.1		3.2	1.5-6.9	

Abbreviations: CI, confidence interval; HR, hazard ratio; MCC+RLN, primary Merkel cell carcinoma with synchronous clinically evident regional nodal disease; MCCUP, Merkel cell carcinoma of unknown primary; SLN, sentinel lymph node.

^a*P* values were calculated with competing risk models for recurrence and disease-specific survival and with Cox regression models for overall survival. *P* values for the nodal group were calculated with Gray's test for cumulative incidence curves of recurrence and disease-specific survival and with a log-rank test for Kaplan-Meier curves of overall survival (see Figs. 2A and 3A, respectively).

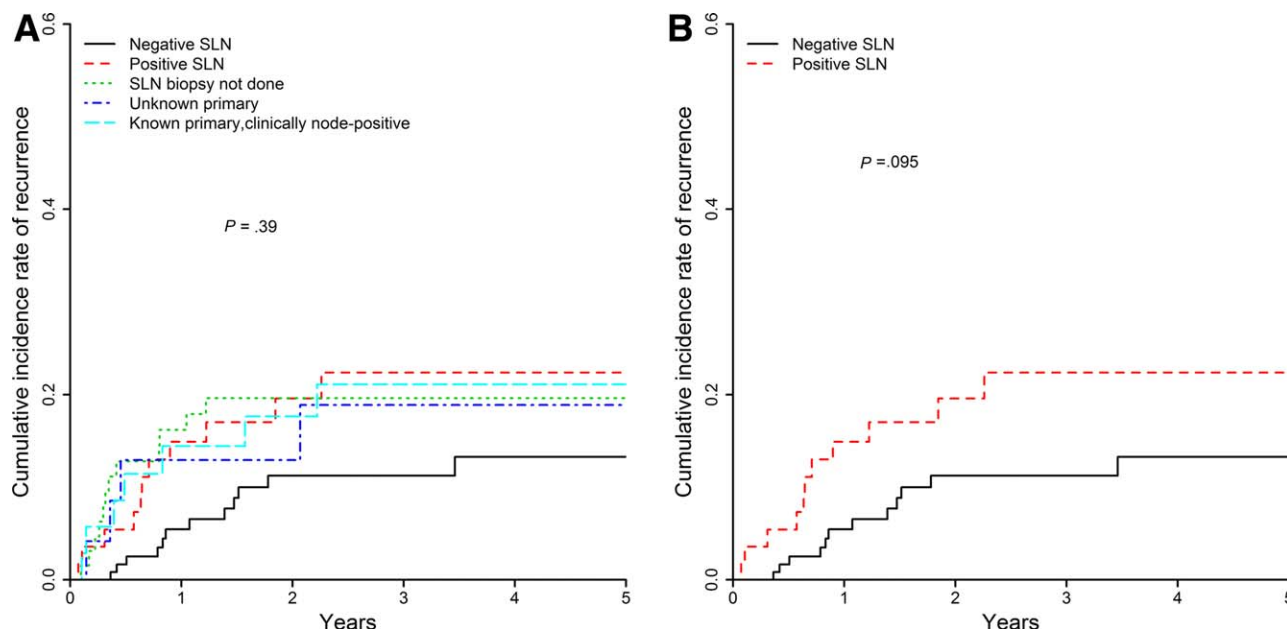


Figure 2. (A) Incidence of recurrence stratified by nodal disease categories. There was no significant difference across the nodal disease categories (*P* = .39). Patients with a negative SLN and those with Merkel cell carcinoma of unknown primary had the fewest recurrences. Patients who did not have an SLN biopsy performed and those with a positive SLN had the highest incidence of recurrence. (B) Incidence of recurrence in SLN-negative and SLN-positive patients. SLN indicates sentinel lymph node.

other causes. In comparison with a primary MCC located in the head and neck or on the extremities, the risk of death was greater for patients with primaries located on

the trunk. The presence of immunosuppression was associated with a worse OS (*P* < .01). The probability of death for immunosuppressed patients was 1.8 times (95%

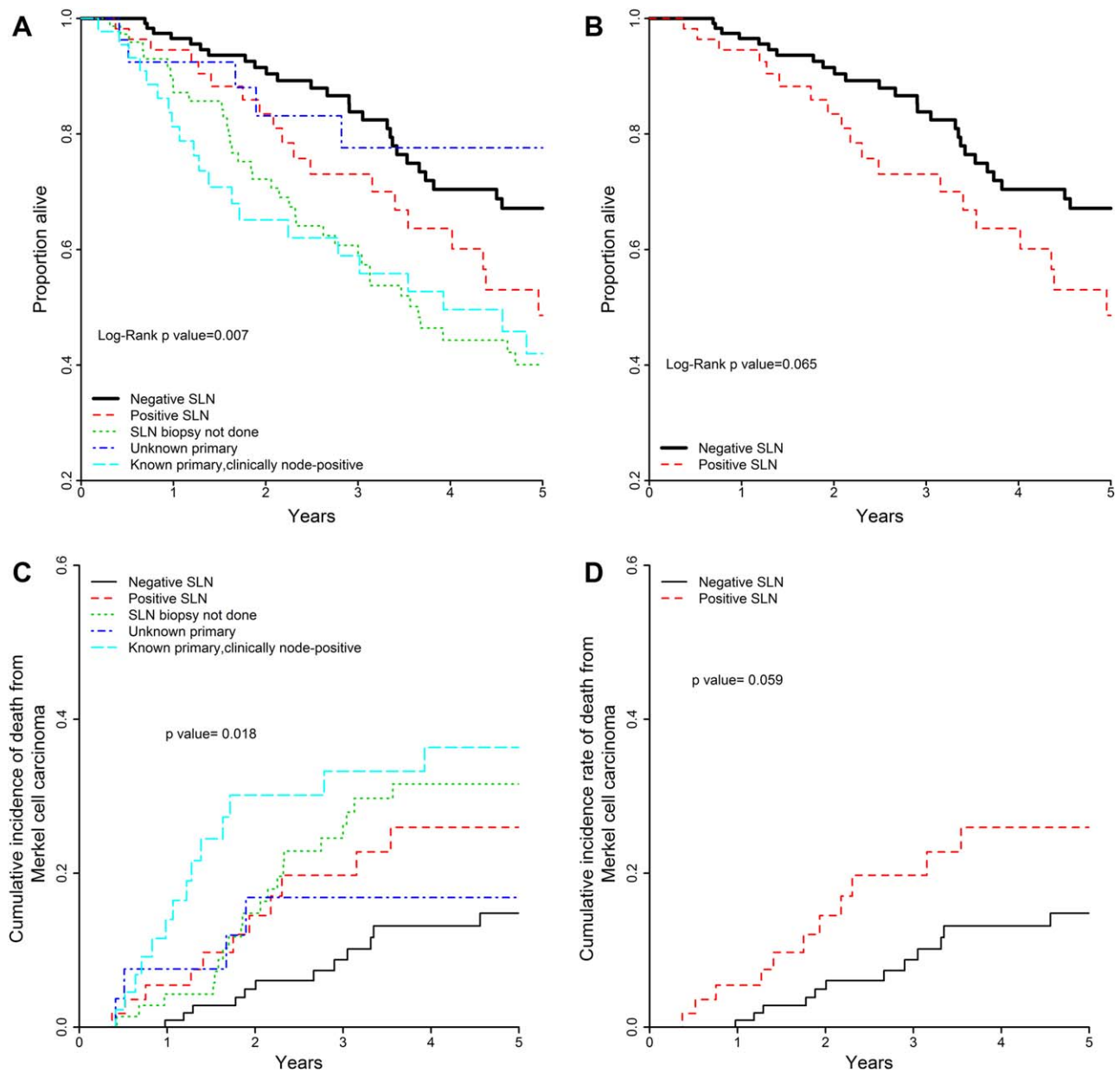


Figure 3. Overall survival and mortality stratified by nodal disease categories. (A) Overall survival stratified by the 5 nodal disease categories. There was a statistically significant difference ($P=.007$) in overall survival between the groups. The best overall survival was seen among patients with MCCUP and those with a negative SLN. (B) Overall survival for SLN-negative and SLN-positive patients ($P=.065$). The median survival was 85 months for the SLN-negative patients and 59.5 months for the SLN-positive patients. (C) Mortality from Merkel cell carcinoma stratified by the 5 nodal categories. Patients with a negative SLN had the best disease-specific survival. The worst disease-specific survival was observed in patients with clinically evident regional nodal disease at presentation. (D) Mortality from Merkel cell carcinoma in SLN-negative and SLN-positive patients ($P=.059$). SLN indicates sentinel lymph node.

CI, 1.2-2.7) greater than that for patients who were not immunosuppressed. Increasing tumor diameter and increasing tumor depth were each independently associated with worse OS ($P=.003$ and $P=.025$, respectively). Increasing age was associated with worse OS ($P<.0001$).

There was no statistically significant association between sex and OS ($P=.27$; Table 3).

There was a statistically significant difference in OS across the 5 nodal status groups (log-rank $P=.007$; Fig. 3A). Patients with a negative SLN biopsy had a median

OS of 85 months (95% CI, 67-120 months) versus 60 months (95% CI, 41-104 months) for patients with a positive SLN biopsy, and this trended toward significance (log-rank $P = .065$; Fig. 3B). In patients presenting with clinically node-positive disease, the median OS was not achieved for MCCUP (lower 25% quartile, 73.2 months), where it was 46.8 months (interquartile range, 20.4-88.8 months) for the patients with a known primary ($P = .038$).

DSS

Male sex and increasing age were both associated with worse DSS ($P = .03$ and $P = .003$, respectively). Both increasing tumor depth and tumor diameter were associated with worse DSS ($P < .01$). The presence of immunosuppression and the location of the primary had no significant impact on DSS ($P = .31$ and $P = .49$, respectively). There was a statistically significant difference in DSS across the 5 nodal status groups (Gray's test, $P = .018$; Fig. 3C). There was a trend toward significance (Gray's test, $P = .059$; hazard ratio, 2.1; 95% CI, 1.0-4.4) in DSS for the SLN-positive group versus the SLN-negative group (Fig. 3D).

The 5-year disease-specific death rate for the SLN-positive group was 26.0%, whereas it was 14.8% for the SLN-negative group. The 5-year disease specific death rate for the MCCUP group was 16.8%, whereas it was 36.3% for the known primary group (Gray's test, $P = .14$).

DISCUSSION

MCC can exhibit aggressive behavior with early and frequent regional nodal involvement, distant metastasis, and high mortality.^{5,18} The development of consistent treatment algorithms and staging guidelines for MCC have been hampered by a number of factors, including the rarity of the disease, inconsistencies in the diagnosis, and incomplete reporting of clinicopathologic features, recurrence, and survival data.¹² Before 2011, most of the clinical data on MCC were derived from small series with fewer than 100 patients. A key exception was a series of 500 patients over a 40-year period reported by the Memorial Sloan Kettering Cancer Center.⁷ The reliability of data from the Surveillance, Epidemiology, and End Results database¹⁹ and the National Cancer Database⁸ has been limited by relatively few patients in comparison with other cutaneous malignancies as well as inconsistent pathologic evaluations and staging and limited data on the specifics of treatment. The design of this study made it possible for us to address some of the salient issues in the clinical management of MCC. These were 1) a demon-

stration of the independent impact of the tumor depth and diameter on the risk of a positive SLN biopsy in MCC, 2) the utility of SLN biopsy as a staging and prognostic tool for OS in MCC, and 3) an examination of the impact of the extent of nodal disease at presentation on survival.

In contrast to melanoma, for which robust data exist about the impact of tumor depth on the risk of a positive SLN and survival, there has been a paucity of similar data for MCC. The staging of MCC has been based on the tumor diameter, clinical extent of nodal involvement, and presence or absence of distant metastatic disease. The significance of tumor depth in MCC is an area of ongoing debate and requires additional investigation. In our 191 patients who underwent SLN biopsy, we found a 31% rate of SLN positivity. This is consistent with data from other published series reporting 19% to 38% positivity rates.^{5,12,20-25} The results of the current study demonstrate that, as in the case of melanoma, tumor depth is an independent prognostic factor for both SLN status and OS. Notably, in our series, depth and diameter were highly correlated, likely because many MCC primary tumors are relatively well circumscribed and sometimes nearly spherical. This is in contrast to melanoma, in which a large superficial spreading component can lead to a large diameter in a relatively thin tumor.

Results from 1 published article²² suggest that SLN biopsy is not indicated for MCC less than 1 cm in diameter. However, multiple other series have been unable to identify a tumor diameter or depth below which the probability of finding a positive SLN is clinically insignificant.^{12,21,23,26} In our study, we could not identify using either diameter or depth any group of patients with less than a 10% estimated chance of a positive SLN biopsy. The 10% threshold (or even 5% threshold) of node positivity is one that is widely used to support SLN biopsy in other cutaneous malignancies and one that we use for older patients with some comorbidities (typical of most MCC patients). Tumors on the lower extremities in our study had the highest risk of a positive SLN in comparison with other locations.

Perhaps the most significant finding of the current study is the clear association between tumor depth and OS and DSS. Tumor depth has promise as a histopathologic indicator, and our results suggest that it can predict SLN status and OS about as well as the diameter. However, it is likely to be more objective, accurate, and reproducible than clinical measurements of the diameter. Because of its rarity and nonspecific clinical appearance, MCC is often not suspected at the time of excisional

biopsy. Consequently, accurate size measurements may not be available. Micrometer measurement of depth is familiar to dermatopathologists and can be readily performed as part of the histopathologic evaluation of all primary MCC specimens. To our knowledge, this is the largest study demonstrating an association between tumor depth and these measures of survival in MCC. Just as for the SLN status, the impact of the tumor depth on survival was independent of the tumor diameter. Other investigators have also demonstrated an association between tumor depth and survival in MCC.^{14-16,27} The results of our study and these previously published data lend support to the role of tumor depth as an independent prognostic factor for MCC. In comparison with the published data on Breslow depth in melanoma and SLN status, these are relatively small numbers. It is our recommendation that tumor depth be routinely examined and reported in the synoptic reporting of primary MCC specimens. This is in keeping with the recommendations of the College of American Pathologists for MCC synoptic reporting.²⁸ With time, this may lead to an improvement in our understanding of the impact of tumor depth as more series or collaborative efforts are published. Knowledge of the relation between tumor depth and diameter and nodal status has implications when one is discussing prognosis and treatment options with a patient in the clinic.

Some authors have reported that the presence of LVI correlates with the likelihood of a positive SLN.^{7,27} LVI was incompletely reported in our patient database because we adopted routine reporting of this pathologic parameter only in recent years. This is one limitation of our study because it was not possible to determine the impact of LVI in the multivariate analyses of nodal status and survival.

Other authors have demonstrated a survival difference in patients with a negative SLN versus a positive SLN.²⁰ In our study, there was a 25-month difference in median survival between SLN-positive and SLN-negative patients. However, this was not statistically significant ($P = .065$). Fields et al,²⁹ in an analysis of 153 patients from Memorial Sloan Kettering Hospital with a median follow-up of 41 months, found no difference in OS between SLN-negative and SLN-positive patients ($P = .89$). This is in contrast to the experience with melanoma, for which the relation between a positive sentinel node and significantly worse OS is well established. One possible explanation for the lack of a statistically significant difference in OS between the SLN-positive and SLN-negative groups in our study is the high rate of cen-

soring in the SLN-negative group due to patients being lost to follow-up or dying of non-MCC causes.

The subgroup of patients for whom SLN biopsy was not performed had the worst OS. The most likely explanation for this is that these patients were not offered SLN biopsy because of their concomitant high-risk comorbid conditions, and indeed many of them died of other causes besides MCC. Survival data for MCC are confounded by the fact that MCC is a disease of the immunosuppressed and the elderly, and many patients will die of comorbid conditions rather than MCC, as was the case in our study, in which 55% of all patients who died of other causes. We continue to recommend routine SLN biopsy for clinically node-negative MCC patients who are medically fit to undergo the procedure.

Among patients presenting with clinically evident lymphadenopathy, patients with MCCUP had improved OS over patients presenting with a known primary and synchronous regional nodal disease. The comparisons of DSS were not significant, likely because of the small sample size. Interestingly, in the current study, patients with MCCUP had among the best OS and DSS along with patients with a negative SLN biopsy. A published study of 218 patients with MCC from Kaiser Permanente Northern California also demonstrated improved survival for patients with MCCUP and a lower risk of distant metastasis in comparison with all other patients.³⁰

In conclusion, increasing tumor depth and increasing tumor diameter are each independently associated with the probability of a positive SLN and worse DSS and OS in MCC. Nodal status is significantly prognostic for DSS and OS. SLN-negative patients had a longer median OS than SLN-positive patients; however, this was not statistically significant. This might possibly have reached significance with additional patients and/or perhaps another outcome measure other than the median OS such as the hazard ratio. SLN biopsy should be considered as part of an effort to further study the question of its prognostic significance in a well-designed multi-institution randomized controlled trial with immunotherapies and/or biologic markers.

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REFERENCES

1. Tokar C. Trabecular carcinoma of the skin. *Arch Dermatol*. 1972; 105:107-110.
2. Moll R, Lowe A, Laufer J, et al. Cytokeratin 20 in human carcinomas. A new histodiagnostic marker detected by monoclonal antibodies. *Am J Pathol*. 1992;140:427-447.
3. Hughes MP, Hardee ME, Cornelius LA, et al. Merkel cell carcinoma: epidemiology, target, and therapy. *Curr Dermatol Rep*. 2014; 3:46-53.
4. Santamaria-Barria JA, Boland GM, Yeap BY, et al. Merkel cell carcinoma: 30-year experience from a single institution. *Ann Surg Oncol*. 2013;20:1365-1373.
5. Allen PJ, Bowne WB, Jaques DP, et al. Merkel cell carcinoma: prognosis and treatment of patients from a single institution. *J Clin Oncol*. 2005;23:2300-2309.
6. Hui AC, Stillie AL, Seel M, et al. Merkel cell carcinoma: 27-year experience at the Peter MacCallum Cancer Centre. *Int J Radiat Oncol Biol Phys*. 2011;80:1430-1435.
7. Fields RC, Busam KJ, Chou JF, et al. Five hundred patients with Merkel cell carcinoma evaluated at a single institution. *Ann Surg*. 2011;254:465-473.
8. Lemos BD, Storer BE, Iyer JG, et al. Pathologic nodal evaluation improves prognostic accuracy in Merkel cell carcinoma: analysis of 5823 cases as the basis of the first consensus staging system. *J Am Acad Dermatol*. 2010;63:751-761.
9. Ortin-Perez J, van Rijk MC, Valdes-Olmos RA, et al. Lymphatic mapping and sentinel node biopsy in Merkel's cell carcinoma. *Eur J Surg Oncol*. 2007;33:119-122.
10. National Comprehensive Cancer Network. Merkel cell carcinoma (version 1.2015). http://www.nccn.org/professionals/physician_gls/pdf/mcc.pdf. Accessed April 4, 2015.
11. Miller SJ, Alam M, Andersen J, et al. Merkel cell carcinoma. *J Natl Compr Canc Netw*. 2009;7:322-332.
12. Fields RC, Busam KJ, Chou JF, et al. Recurrence and survival in patients undergoing sentinel lymph node biopsy for Merkel cell carcinoma: analysis of 153 patients from a single institution. *Ann Surg Oncol*. 2011;18:2529-2537.
13. Koljonen V, Tukiainen E, Haglund C, et al. Proliferative activity detected by Ki67 correlates with poor outcome in Merkel cell carcinoma. *Histopathology*. 2006;49:551-553.
14. Goldberg SR, Neifeld JP, Frable WJ. Prognostic value of tumor thickness in patients with Merkel cell carcinoma. *J Surg Oncol*. 2007;95:618-622.
15. Sandel HDT, Day T, Richardson MS, et al. Merkel cell carcinoma: does tumor size or depth of invasion correlate with recurrence, metastasis, or patient survival? *Laryngoscope*. 2006;116:791-795.
16. Andea AA, Coit DG, Amin B, et al. Merkel cell carcinoma: histologic features and prognosis. *Cancer*. 2008;113:2549-2558.
17. Llombart B, Monteagudo C, Lopez-Guerrero JA, et al. Clinicopathological and immunohistochemical analysis of 20 cases of Merkel cell carcinoma in search of prognostic markers. *Histopathology*. 2005; 46:622-634.
18. Heath M, Jaimes N, Lemos B, et al. Clinical characteristics of Merkel cell carcinoma at diagnosis in 195 patients: the AEIOU features. *J Am Acad Dermatol*. 2008;58:375-381.
19. Albores-Saavedra J, Batich K, Chable-Montero F, et al. Merkel cell carcinoma demographics, morphology, and survival based on 3870 cases: a population based study. *J Cutan Pathol*. 2010;37:20-27.
20. Gupta SG, Wang LC, Penas PF, et al. Sentinel lymph node biopsy for evaluation and treatment of patients with Merkel cell carcinoma: the Dana-Farber experience and meta-analysis of the literature. *Arch Dermatol*. 2006;142:685-690.
21. Schwartz JL, Griffith KA, Lowe L, et al. Features predicting sentinel lymph node positivity in Merkel cell carcinoma. *J Clin Oncol*. 2011; 29:1036-1041.
22. Stokes JB, Graw KS, Dengel LT, et al. Patients with Merkel cell carcinoma tumors ≤ 1.0 cm in diameter are unlikely to harbor regional lymph node metastasis. *J Clin Oncol*. 2009;27:3772-3777.
23. Bajetta E, Celio L, Platania M, et al. Single-institution series of early-stage Merkel cell carcinoma: long-term outcomes in 95 patients managed with surgery alone. *Ann Surg Oncol*. 2009;16:2985-2993.
24. Warner RE, Quinn MJ, Hruby G, et al. Management of Merkel cell carcinoma: the roles of lymphoscintigraphy, sentinel lymph node biopsy and adjuvant radiotherapy. *Ann Surg Oncol*. 2008;15:2509-2518.
25. Hill AD, Brady MS, Coit DG. Intraoperative lymphatic mapping and sentinel lymph node biopsy for Merkel cell carcinoma. *Br J Surg*. 1999;86:518-521.
26. Sarnaik AA, Zager JS, Cox LE, et al. Routine omission of sentinel lymph node biopsy for Merkel cell carcinoma ≤ 1 cm is not justified. *J Clin Oncol*. 2010;28:e7.
27. Lim CS, Whalley D, Haydu LE, et al. Increasing tumor thickness is associated with recurrence and poorer survival in patients with Merkel cell carcinoma. *Ann Surg Oncol*. 2012;19:3325-3334.
28. Rao P, Balzer BL, Lemos BD, et al. Protocol for the examination of specimens from patients with Merkel cell carcinoma of the skin. *Arch Pathol Lab Med*. 2010;134:341-344.
29. Fields RC, Busam KJ, Chou JF, et al. Recurrence after complete resection and selective use of adjuvant therapy for stage I through III Merkel cell carcinoma. *Cancer*. 2012;118:3311-3320.
30. Asgari MM, Sokil MM, Warton EM, et al. Effect of host, tumor, diagnostic, and treatment variables on outcomes in a large cohort with Merkel cell carcinoma. *JAMA Dermatol*. 2014;150:716-723.